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Exploratory Analysis of UNSW-NB15 Dataset for Detecting Malicious Network Traffic

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Cyber-attacks against individuals, businesses, and nations hit headlines almost every day as computers, mobile communication devices, IoT devices and networks have become an integral part of our life. Unfortunately, these devices can be attacked by network intruders and cyber criminals with malicious intent. Powerful security defence mechanisms involving intrusion detection systems are essential for detecting such cyber-attacks. It is desirable that the machine learns itself the behaviour of incoming packets and detect any suspicious or anomalous behaviour. Enormous benefits of deep learning can also be applied in cybersecurity domain.

This article is written with the goal to introduce readers aiming to begin research in the area of cyber security domain, specifically detecting malicious activities using machine learning. The article shares few insights from the exploratory analysis of a benchmark dataset called UNSW-NB15 [1].

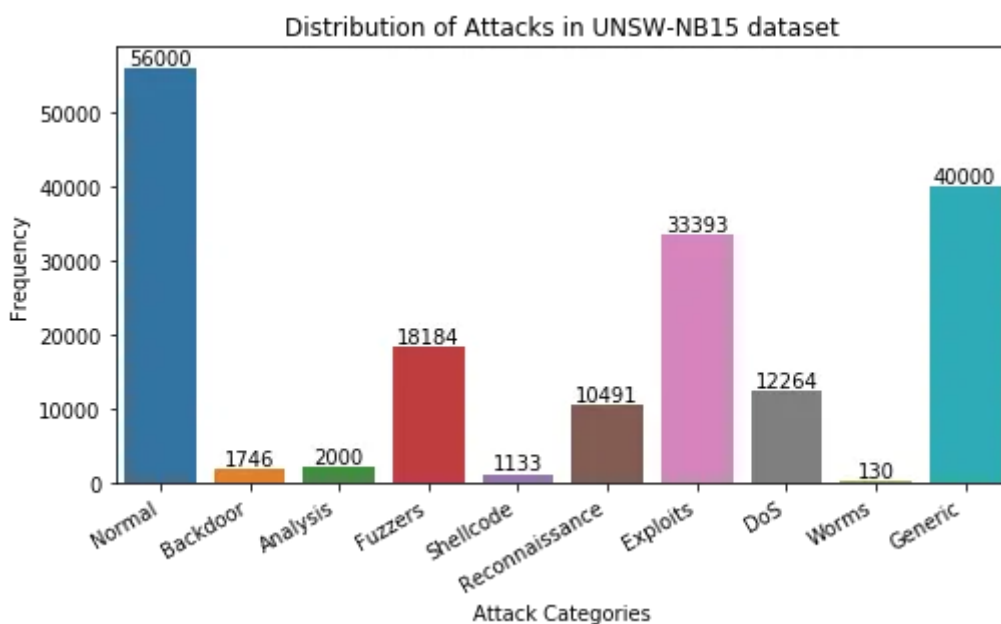
Datasets: NSL KDD v/s UNSW-NB15

The most important constituent of any machine learning project is data. Most existing research for network intrusion detection has been done using the ubiquitous NSL-KDD dataset [2] which was created long back in 2000. The network

data and attack pattern have drastically changed since then. Hence, to correctly gauge the performance of machine learning or deep learning techniques, systematic evaluation should be done on recent datasets. The UNSW-NB15 dataset bridges this gap and covers a variety of modern attacks specified by McAfee threat report 2018[3]. The following table compares the NSL-KDD and UNSW-NB15 datasets based on various parameters:

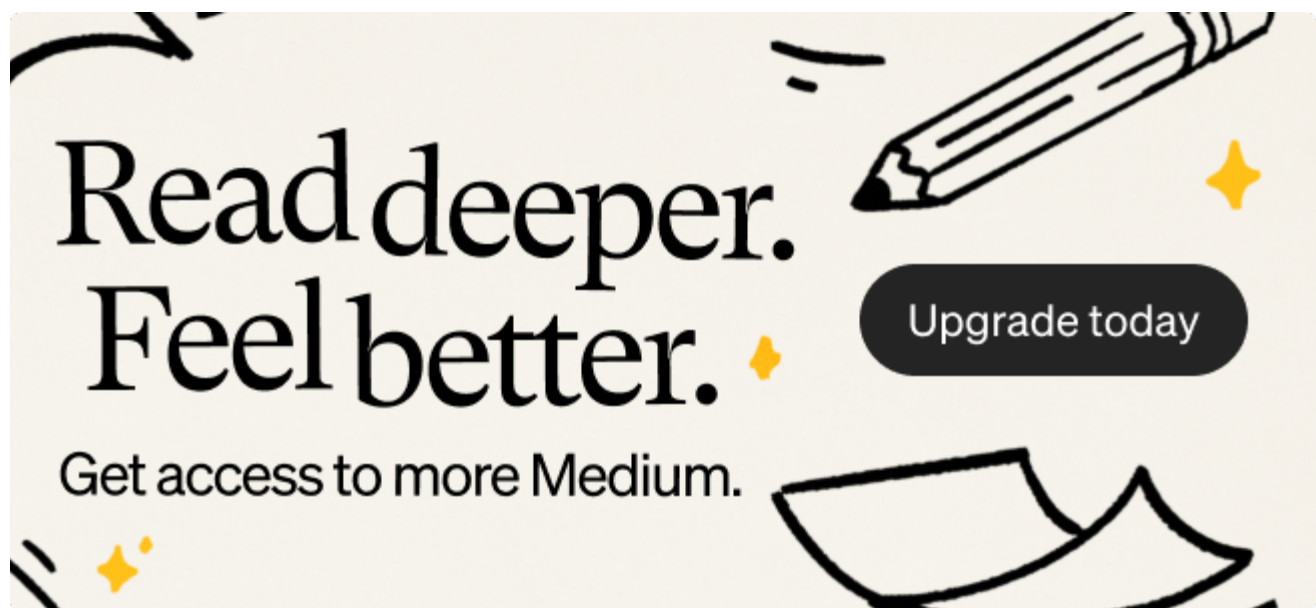
Parameters	NSL KDD	UNSW- NB 15
Year of Creation	2000	2015
Modern Network Traffic and Attacks	No	Yes
Attack Traffic Generation	Simulated	IXIA perfect storm tool
Duration of Data Collected	7 weeks	16 hours &15 hours
Feature Extraction Tools	<u>BroIDS</u>	<u>BroIDS</u> , Argus
# Features	41	47
# Attack Categories	4	9
Publicly Available	Yes	Yes
Labelled dataset	Yes	Yes

For a visual analysis of the dataset, the Matplotlib Python library was used to observe the distribution of various attacks in the UNSW-NB15 dataset. The frequency of occurrence of various attack categories is shown in the following figure:

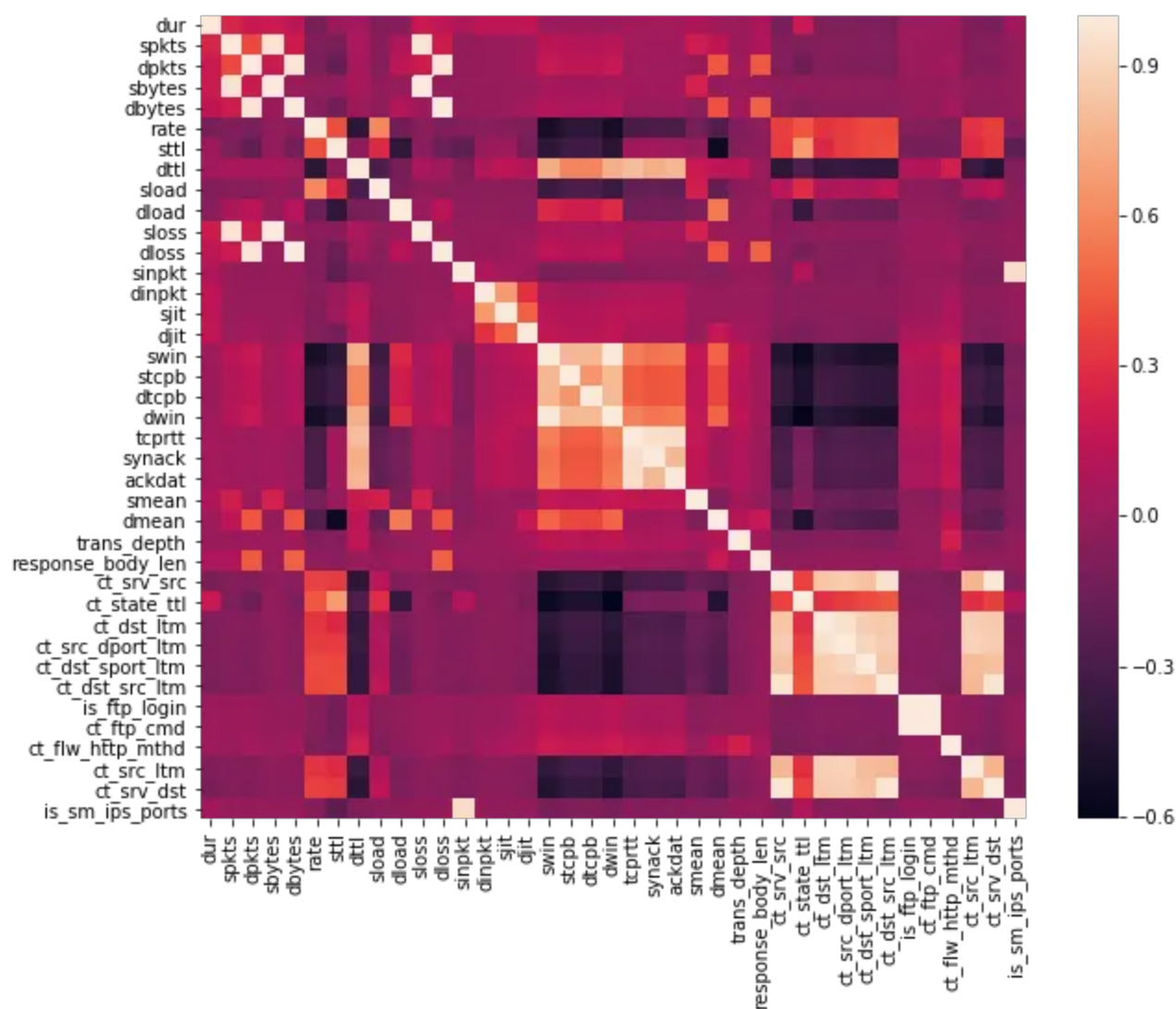


As compared to the NSL-KDD dataset, these attack categories represent the current cyber scenario. In addition, it is observed that the distribution of attacks is highly imbalanced which depicts a real-world situation. This is a very important issue to be considered when applying any machine learning algorithm; the imbalance ratio should be considered when learning the patterns as well as when evaluating the performance of the classifier. For example, a cost-sensitive machine learning algorithm can be deployed to learn the attack patterns where the cost reflects the imbalance ratio.

Feature Correlation:



To determine the correlation between various network features of the UNSW-NB15 dataset, Pearson correlation coefficient was measured and the corresponding scatter plot of correlation coefficients is shown in the following figure.

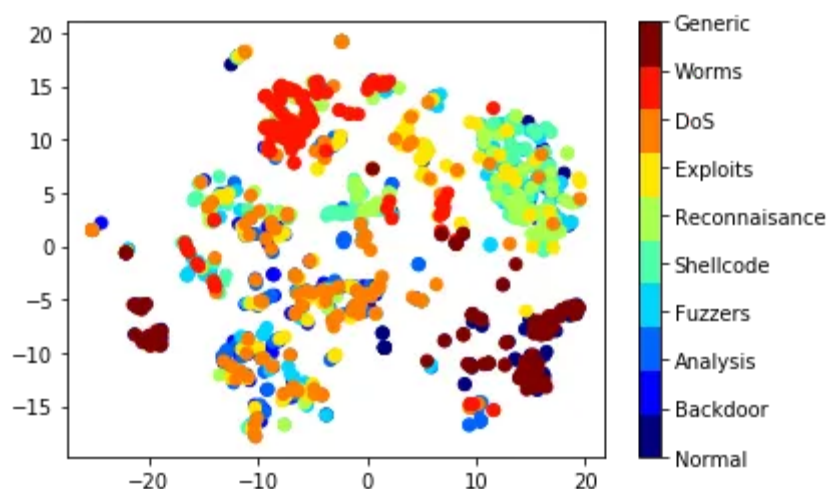


Pearson's correlation coefficient is used to determine the relationship between two features. It gives the measure of strength of correlation between two variables. The value of coefficient ranges between -1 to +1, where +1 means that they are highly correlated and -1 means that there is a negative correlation and 0 means no correlation. The dataset consists of various network traffic features. We observe from the scatter plot that there are few features that are completely correlated so keeping both of them will be redundant for any learning algorithm. For example, the correlated feature pairs are {sloss, sbytes}, {dloss, dbytes}, {is_ftp_login, ct_ftp_cmd}; it would be prudent to remove any feature from these pairs to avoid redundancy. Another example is the feature pair sbytes and spackets having a high correlation since as spackets will increase sbytes also increases. Thus, it would be beneficial to apply some dimensionality reduction techniques such as Principal

Component Analysis (PCA) or Auto-Encoder and retain only the most relevant and uncorrelated features.

t-SNE Plot:

Due to high feature dimensionality (41) of the UNSW-NB15 dataset, visually exploring the data is a challenging task. Thus, we use a technique known as t-SNE (t-distributed Stochastic Neighbor Embedding) to project the features to a lower dimensional space by minimizing the Kullback-Leibler divergence between the distribution of higher dimensional features and distribution of lower dimensional projected features. t-SNE thus attempts to find patterns in the data by identifying observed clusters based on similarity of data points with multiple features. The t-sne scatter plot of UNSW-NB15 dataset projected onto two dimensions is as shown:



From the t-sne plot, it can be observed that there is a huge overlap between various attacks that suggests that UNSW- NB15 is a complex dataset to deal with for identifying specific attack categories.

Conclusion:

Research in the area of detecting malicious network traffic is a hot topic since the problem is not yet completely solved. Most commercial techniques still rely on signature-based mechanisms such as Snort. State-of-the-art research work in this area uses machine learning and deep learning techniques; however, most work still uses the older dataset of NSL-KDD for evaluation. We encourage researchers to

adopt the recent UNSW-NB15 dataset that is made freely available to the public by the authors of [1], since it represents modern network scenario as well as the attack categories and distribution as observed in real-life networks.

References:

1. Moustafa, Nour, and Jill Slay. "UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)." *2015 military communications and information systems conference (MilCIS)*. IEEE, 2015.

2. <https://www.unb.ca/cic/datasets/nsl.html>

3. <https://www.mcafee.com/enterprise/en-us/assets/reports/rp-quarterly-threats-dec-2018.pdf>

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